Prelims Introductory Calculus MT 2026: Sheet 8

1. Find and classify all the critical points of each of the following functions

   (a)
   $$f(x,y) = 6x^2 - 2x^3 + 3y^2 + 6xy,$$

   (b)
   $$f(x,y) = xy(x^2 + y^2 - 1),$$

   (c)
   $$f(x,y) = \sin^2 x + \sin^2 y - \cos^2 x \cos^2 y.$$

2. Let
   $$f(x,y) = x^3 + x^2 + 2\alpha xy + y^2 + 2\alpha x + 2y,$$
   where α is a positive constant. Find and classify all the critical points of $f(x,y)$ in each of the followng cases:
   (a) when $\alpha > 1$, (b) when $0 < \alpha < 1$.

3. Find the volume of the largest cuboid (i.e. box) with edges parallel to the axes, inscribed in the ellipsoid
   $$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1.$$

4. Heron's formula for the area A of a triangle with sides of length $a, b, c > 0$ is
   $$A = \sqrt{s(s-a)(s-b)(s-c)}, \quad \text{where } s = \frac{a+b+c}{2}$$
   and hence $2s$ is the perimeter of the triangle.
   Use the method of Lagrange multipliers to answer the following:
   (a) Show that, for a given fixed perimeter P of the triangle, the area is maximised when the triangle is equilateral. Determine this maximum area in terms of P.
   (b) Find the maximum area of a right-angled triangle with given fixed perimeter P, in terms of P.

5. Use the method of Lagrange multipliers to find the shortest distance from the origin to the line of intersection of the planes $2x + y - z = 1$ and $x - y + z = 2$.
   Optional (for those taking Geometry): Repeat the calculation using geometric methods.

6. Find all the critical points (x, y, z) of the function
   $$f(x,y,z) = x^2 + y^2 + z^2$$
   subject to the constraints that
   $$x + y - z = 0 \qquad \text{and} \qquad \frac{x^2}{4} + \frac{y^2}{5} + \frac{z^2}{25} = 1.$$
   Detemine the value of f at these critical points. Intrepret the critical points in terms of a geometry problem.